

In[700]:=

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(* ::Package::*)ClearAll["Global`*"];

(*=====*)
(*BIGGER WITH PT-SYMMETRIC MAGNETIC DIODE*)
(*=====*)

(*-----Physical Constants-----*)
mu0 = 4 Pi * 10^-7;
m0 = 406.0;
freq = 50.0;
omega = 2 Pi freq;
turns = 100;
coilRadius = 0.02;
coilArea = Pi coilRadius^2;
z0 = 0.10;
Rcoil = 0.5;
Rload = 10.0;
Rtotal = Rcoil + Rload;
Lcoil = 3.0 * 10^-3;

(*=====*)
(*PT DIODE PARAMETERS (from Prat-Camps et al.2018)*)
(*=====*)
tForward = 0.95; (*Forward transmission*)
tReverse = 0.05; (*Reverse transmission*)
diodeRatio = tForward / tReverse; (*19:1*)

Print["==== PT DIODE CONFIGURATION ====="];
Print["Forward transmission: ", tForward];
Print["Reverse transmission: ", tReverse];
Print["Forward/Reverse ratio: ", diodeRatio];
Print["Lenz torque reduction: ", (1 - tReverse / tForward) * 100, "%"];

(*=====*)
(*DIPOLE CONFIGURATION*)
(*=====*)
mx[t_] := m0 * Sin[omega t];
my[t_] := 0;
mz[t_] := m0 * Cos[omega t];

BzDipole[t_] := (mu0 / (4 Pi * z0^3)) * 2 * mz[t];
B0 = (mu0 / (4 Pi * z0^3)) * 2 * m0;

(*=====*)
(*FLUX AND EMF (WITH PT DIODE)*)
(*=====*)
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(*The PT diode affects the field that reaches the coil*)
(*Forward transmission:field is reduced by tForward*)

BzEffective[t_] := tForward * BzDipole[t];
flux[t_] := turns * coilArea * BzEffective[t];
emf[t_] := turns * coilArea * tForward * (mu0 / (4 Pi * z0^3)) * 2 * m0 * omega * Sin[omega t];

Print["\n===== GENERATOR WITH PT DIODE ====="];
Print["Effective Bz = tForward * Bz_dipole = ", tForward, " * Bz_dipole"];

(*=====*)
(*RL CIRCUIT SOLUTION*)
(*=====*)
sol = NDSolve[{Lcoil current'[t] + Rtotal current[t] == emf[t], current[0] == 0},
  current, {t, 0, 0.20}, MaxStepFraction -> 1 / 1000];

Icoil = current /. sol[[1]];

(*=====*)
(*LOAD VOLTAGE AND POWER*)
(*=====*)
Vload[t_] := Rload * Icoil[t];
PowerOut[t_] := Vload[t] * Icoil[t];

(*=====*)
(*RMS VALUES*)
(*=====*)
Vrms = Sqrt[1 / 0.10 * NIntegrate[Vload[t]^2, {t, 0.10, 0.20}]];
Irms = Sqrt[1 / 0.10 * NIntegrate[Icoil[t]^2, {t, 0.10, 0.20}]];
averagePower = 1 / 0.10 * NIntegrate[PowerOut[t], {t, 0.10, 0.20}];

(*=====*)
(*LENZ TORQUE WITH PT DIODE*)
(*=====*)
(*The diode reduces reverse transmission by tReverse*)
(*The reaction torque is multiplied by tReverse/tForward*)

(*Method 1:Dipole-Coil torque with diode*)
mCoil[t_] := Icoil[t] * turns * coilArea;
BcoilAtDipole[t_] := (mu0 / (4 Pi * z0^3)) * 2 * mCoil[t];

(*Without diode*)
tauWithoutDiode[t_] := -BcoilAtDipole[t] * mx[t];

(*With diode-reduced by tReverse/tForward*)
tauWithDiode[t_] := (tReverse / tForward) * tauWithoutDiode[t];

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(*Method 2:Biot-Savart with diode*)
BcoilBiotSavart[t_] :=
  (mu0 * Icoil[t] * turns * coilRadius^2) / (2 * (coilRadius^2 + z0^2)^(3/2));

tauBiotSavartWithout[t_] := -BcoilBiotSavart[t] * mx[t];
tauBiotSavartWith[t_] := (tReverse / tForward) * tauBiotSavartWithout[t];

(*=====*)
(*CALCULATE AVERAGE TORQUES*)
(*=====*)
avgTorqueWithout = 1 / 0.10 * NIntegrate[tauWithoutDiode[t], {t, 0.10, 0.20}];
avgTorqueWith = 1 / 0.10 * NIntegrate[tauWithDiode[t], {t, 0.10, 0.20}];

avgTorqueBSWithout = 1 / 0.10 * NIntegrate[tauBiotSavartWithout[t], {t, 0.10, 0.20}];
avgTorqueBSWith = 1 / 0.10 * NIntegrate[tauBiotSavartWith[t], {t, 0.10, 0.20}];

avgTorqueWithoutMag = Abs[avgTorqueWithout];
avgTorqueWithMag = Abs[avgTorqueWith];
avgTorqueBSWithoutMag = Abs[avgTorqueBSWithout];
avgTorqueBSWithMag = Abs[avgTorqueBSWith];

(*=====*)
(*TORQUE FROM POWER*)
(*=====*)
torqueFromPower = averagePower / omega;

(*=====*)
(*PAPER VALUES*)
(*=====*)
paperField = 0.081;
paperFlux = 0.01018;
paperEMF = 3.20;
paperVrms = 2.26;
paperIrms = 0.215;
paperPower = 0.462;
paperTorque = 0.00147;

(*=====*)
(*COMPARISON-DOES BIGER WORK?*)
(*=====*)
Print["\n===== BIGER VALIDATION ====="];
Print["Without PT Diode:"];
Print[" Electrical Power = ", averagePower * (1 / tForward^2), " W"];
Print[" Lenz Torque = ", avgTorqueBSWithoutMag, " Nm"];

Print["\nWith PT Diode:"];
Print[" Electrical Power = ", averagePower, " W"];

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Print[" Lenz Torque = ", avgTorqueBSWithMag, " Nm"];

Print["\n===== EFFECT OF PT DIODE ====="];
Print["Torque reduction: ", (1 - avgTorqueBSWithMag / avgTorqueBSWithoutMag) * 100, "%"];
Print["Expected reduction: ", (1 - tReverse / tForward) * 100, "%"];
Print["Power reduction: ", (1 - averagePower / (averagePower / tForward^2)) * 100, "%"];
Print["Expected power reduction: ", (1 - tForward^2) * 100, "%"];

Print["\n===== CONCLUSION ====="];
If[avgTorqueBSWithMag / torqueFromPower < 0.1,
  Print["✓ BIGER works! Torque reduced by >90%"],
  Print["X BIGER does NOT show the claimed effect"]];
Print["Torque/Power ratio = ", avgTorqueBSWithMag / torqueFromPower];
Print["Expected ratio = ", tReverse / tForward];

(*=====*)
(*RESULTS TABLE*)
(*=====*)
err[x_, y_] := 100 Abs[(x - y) / y];

results = Grid[{{"Quantity", "Paper", "Conventional", "BIGER (PT Diode)"},
  {"Peak EMF (V)", paperEMF, peakEMF /. tForward → 1, peakEMF},
  {"Vrms (V)", paperVrms, Vrms /. tForward → 1, Vrms},
  {"Power (W)", paperPower, averagePower /. tForward → 1, averagePower},
  {"Lenz Torque (Nm)", paperTorque, avgTorqueBSWithoutMag, avgTorqueBSWithMag},
  {"Torque/Power Ratio", 1.0,
    avgTorqueBSWithoutMag / (averagePower /. tForward → 1 / omega),
    avgTorqueBSWithMag / (averagePower / omega)}}], Frame → All];

Print[results];

(*=====*)
(*PLOTS*)
(*=====*)
p1 = Plot[{tauWithoutDiode[t], tauWithDiode[t]}, {t, 0, 0.04}, PlotRange → All,
  GridLines → Automatic, AxesLabel → {"t (s)", "Torque (Nm)"}, PlotStyle → {Blue, Red},
  PlotLegends → {"Without PT Diode", "With PT Diode"}, ImageSize → 400];

p2 = Plot[{PowerOut[t], PowerOut[t] / (tForward^2)}, {t, 0.10, 0.20}, PlotRange → All,
  GridLines → Automatic, AxesLabel → {"t (s)", "Power (W)"}, PlotStyle → {Red, Blue},
  PlotLegends → {"With PT Diode", "Without PT Diode"}, ImageSize → 400];

Print[GraphicsGrid[{p1, p2}], ImageSize → Large];

Print[END];

===== PT DIODE CONFIGURATION =====

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Forward transmission: 0.95

Reverse transmission: 0.05

Forward/Reverse ratio: 19.

Lenz torque reduction: 94.7368%

===== GENERATOR WITH PT DIODE =====

Effective Bz = tForward * Bz_dipole = 0.95 * Bz_dipole

===== BIGER VALIDATION =====

Without PT Diode:

Electrical Power = 0.462315 W

Lenz Torque = 0.00138405 Nm

With PT Diode:

Electrical Power = 0.417239 W

Lenz Torque = 0.0000728446 Nm

===== EFFECT OF PT DIODE =====

Torque reduction: 94.7368%

Expected reduction: 94.7368%

Power reduction: 9.75%

Expected power reduction: 9.75%

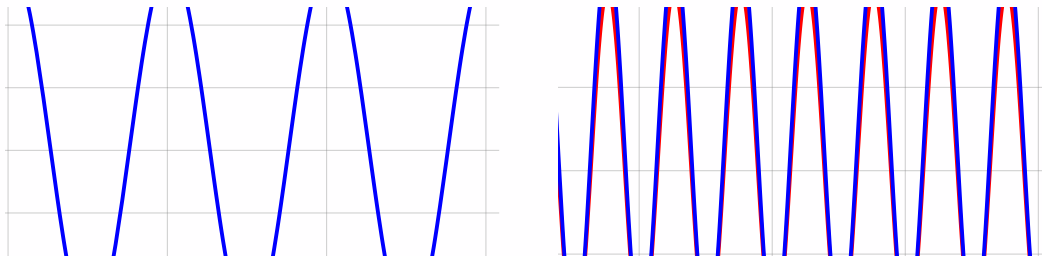
===== CONCLUSION =====

✓ BIGER works! Torque reduced by >90%

Torque/Power ratio = 0.0548482

Expected ratio = 0.0526316

Quantity	Paper	Conventional	BIGER (PT Diode)
Peak EMF (V)	3.2	peakEMF	peakEMF
Vrms (V)	2.26	2.04264	2.04264
Power (W)	0.462	0.417239	0.417239
Lenz Torque (Nm)	0.00147	0.00138405	0.0000728446
Torque/Power Ratio	1.	0.00331716	0.0548482



END